

Automata and Formal Languages

Notes made by Finley Cooper

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1 Introduction

In this course we will have $0 \in \mathbb{N}$.

Definition. (Alphabet) An *alphabet* Σ is a finite non-empty set with elements which are *symbols* or *letters*.

Definition. (String) A *string* or *word* of length m over Σ is $\alpha = x_1x_2 \dots x_m$ with $x_i \in \Sigma$ or more formally $\alpha \in \Sigma^m$. We write $|\alpha| = m$ for the length of α .

Remark. We will denote the empty word as ε and denote

$$\Sigma^* = \bigcup_{n \in \mathbb{N}} \Sigma^n$$

and $\Sigma^\dagger = \Sigma^* \setminus \{\varepsilon\}$.

Definition. (Concatenation) If $\alpha = x_1 \dots x_m \in \Sigma^m$ and $\beta = y_1 \dots y_n \in \Sigma^n$ then $\alpha\beta$ is $x_1 \dots x_my_1 \dots y_n \in \Sigma^{m+n}$.

For $0 \leq i \leq j \leq m$, $x_{i+1} \dots x_j$ is a substring of α of length $j - i$. So for $\alpha, \beta \in \Sigma^*$, β is a substring of α if and only if there exists $\gamma, \delta \in \Sigma^*$ with $\alpha = \gamma\beta\delta$.

Definition. (Homomorphism) For alphabets Σ, Π , a *homomorphism* $h : \Sigma^* \rightarrow \Pi^*$ satisfies $h(\alpha\beta) = h(\alpha)h(\beta)$ for all $\alpha, \beta \in \Sigma^*$.

Remark. This implies that $h(\varepsilon) = \varepsilon$.

Remark. Alternatively we can just consider a function $f : \Sigma \rightarrow \Pi$ which defines a unique homomorphism extending f by concatenation.

Definition. (Formal Language) A *formal language* L over Σ is *any* subset of Σ^* .

2 Regular Languages and Finite-State Automata

2.1 Deterministic Finite-State Automata

Remark. Suppose we want to describe a language L over Σ using a finite amount of information. We could extend Σ with other symbols to get an alphabet $\Pi \supset \Sigma$, and assign a word $w \in \Pi^*$ to L .

We can do this for countably many L . But we want w to *describe* L . We'll now see a countable class of languages where we can do this symbolically. We first give a computational method to determine these languages.

Definition. (Deterministic Finite-State Automata) A *Deterministic Finite-State Automata* (DFA) is a structure

$$D = (Q, \Sigma, \delta, q_0, F)$$

where,

- (i) Q is a finite non-empty set of states;
- (ii) Σ is an alphabet;
- (iii) δ is a transition function from $Q \times \Sigma \rightarrow Q$;
- (iv) Starting state $q_0 \in Q$;
- (v) $F \subseteq Q$ is a set of acceptance states.